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Gaze behavior and decision-making in simulated defensive situations for baseball fielders using a head-mounted display

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ABSTRACT

Decision-making in baseball fielding requires fast and accurate assessments of multiple visual cues, including ball trajectory and runner movement. This study investigated the relationship between gaze behavior and decision-making performance during defensive plays, specifically focusing on the selection of the appropriate base to throw to. Twenty collegiate baseball fielders participated in a video-based decision-making task using a head-mounted display. Participants viewed first-person perspective videos simulating a defensive situation with one out and a runner on third base and were asked to decide whether to throw to first base or home plate. Decisions were made by pressing a button during the interval between bat–ball contact and ball capture, reflecting the real-time constraints of actual gameplay. Decision accuracy, decision time, and response bias were measured. Eye-tracking data were also collected to analyze gaze behavior such as the number of fixations and gaze transitions between areas of interest. Results revealed that decision accuracy peaked when participants made an intermediate number of fixations per trial. Gaze transition patterns showed that participants initially fixated on the ball in most trials. Trials in which participants gazed at home plate, runner, or other areas were associated with significantly longer decision times. Moreover, trials involving gaze transitions to other areas were associated with significantly lower decision accuracy. Regarding response bias, participants who spent more time viewing the ball were more likely to make assertive throw decisions by choosing home plate, whereas those who made more gaze transitions between areas of interest tended to make more conservative choices, such as selecting first base. These findings suggest that efficient gaze behavior, characterized by an optimal number of fixations and appropriate allocation of gaze, are crucial for making fast and accurate decisions during defensive plays in baseball.

1. Introduction

In ball sports, players are required to make accurate decisions under time constraints based on various environmental cues, such as the ball, teammates, and opponents (Roca et al., 2013). In baseball, earlier studies have primarily examined decision-making and visual search strategies in batting situations as highlighted in a review (Toole & Fogt, 2021). In contrast, few studies have investigated decision-making in defensive situations, where fielders must quickly decide which base to throw to after catching the ball to complete a play successfully (Hepler & Feltz, 2012). The present study focuses on decision-making related to the selection of the appropriate base to throw to during a defensive play in baseball under strict time constraints.

Numerous studies have investigated the decision-making processes of athletes in ball sports. Many of these studies have reported that skilled players make faster and more accurate decisions than their less skilled counterparts (e.g., Helsen & Pauwels, 1993; Piras et al., 2014; Takeuchi & Inomata, 2009; Vaeyens et al., 2007b). More recently, several studies on perceptual-cognitive abilities, predictive skills, and decision-making in sports have employed signal detection theory (Green & Swets, 1966) as an analytical framework, in addition to evaluating decision-making speed and accuracy (Bruce et al., 2012; Cañal-Bruland & Schmidt,

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2009; Cañal-Bruland et al., 2015; Catteeuw et al., 2009; Gray, 2010; Hancock & Ste-Marie, 2013; Jackson et al., 2018, 2020; Kikumasa & Kokubu, 2018; Mori & Ono, 2023). Signal detection theory treats external stimuli as either noise or signal and distinguishes between signal detection sensitivity (*d'*), which refers to the ability to discriminate between noise and signal, and response bias (β), which represents a tendency to favor one type of response over another. Research using signal detection theory has shown that experts exhibit higher signal detection sensitivity (*d'*) than non-experts (Cañal-Bruland & Schmidt, 2009; Gray, 2010), and that skilled players develop specific likelihood ratios or response biases (β) based on their playing experience, which differ from those of less skilled players (Cañal-Bruland & Schmidt, 2009).

In research on baseball, Kikumasa & Kokubu (2018) investigated the decision-making processes of catchers using video clips depicting a bunt to the pitcher with a runner on first base and no outs, filmed from the catcher's perspective. Participants were asked to decide whether to instruct the pitcher to throw to first or second base by pressing a button during the interval between bat–ball contact and the pitcher's catch. The results showed that catchers exhibited higher signal detection sensitivity than fielders and novices in judging whether the runner would be out or safe. Furthermore, baseball players (both catchers and fielders) demonstrated a higher response bias (β) toward directing throws to first base compared to novices. These findings indicate that baseball players make situation-specific decisions. However, the relationship between gaze behavior and decision-making has not yet been examined in defensive situations in baseball.

One factor associated with the superior decision-making of skilled players is their gaze strategy. Previous studies investigating gaze strategies in sports have examined the characteristics of skilled players by comparing across different levels of competition (Afonso et al., 2012; Nagano et al., 2004; Natsuhara et al., 2020; Piras et al., 2014; Ripoll et al., 1995; Rivilla-Garcia et al., 2013; Roca et al., 2011; Vaeyens et al., 2007b; Williams & Davids, 1998) and different situational conditions (Piras et al., 2016; Savelsbergh et al., 2005; Vaeyens et al., 2007a). There have been mixed findings regarding gaze strategies associated with skilled performance (Ziv, 2025). Mann et al. (2007) reported in a meta-analysis that experts tend to use fewer but longer fixations. Their study also noted that the relationship between level of expertise and perceptual-cognitive skills was moderated by factors such as the type of sport, the presented stimuli, and the research paradigm employed. In fact, excessive fixation on the ball alone has even been linked to poor performance in soccer goalkeepers (Piras & Vickers, 2011). It has been reported that elite handball goalkeepers exhibit a greater number of fixations than their less-skilled counterparts (Rivilla-García et al., 2013). Taken together, it is postulated that neither very few nor excessively many fixations are optimal. In other words, decision accuracy is expected to be maximized when players make a moderate number of fixations.

In terms of gaze transitions, Vaeyens et al. (2007b) found that soccer players with superior decision-making skills spent more time fixating on the ball carrier and alternated gaze more frequently between the ball carrier and other relevant areas compared to those with poorer decision-making skills. Expert players also employ strategies such as visual pivots in order to briefly gaze at task-relevant areas and rapidly gather information (Piras & Vickers, 2011; Ripoll et al., 1995). Based on these findings, we expected that gaze directed toward task-irrelevant areas would impair decision-making performance in the present study.

As mentioned above, many previous studies have compared gaze strategies between groups. However, even among skilled players, there are individual differences in decision accuracy and response bias (β or *c*), and these differences may be related to their gaze strategies. A few studies on baseball fielders' decisions regarding which base to throw to have examined factors such as the role of self-efficacy (Hepler & Feltz, 2012) and have used virtual reality to simulate outfielders' throwing decisions after catching fly balls, employing a gaze-based targeting interface (Lee et al., 2024). However, no studies to date have specifically investigated the gaze strategies underlying fielders' decisions. Therefore, the present study investigated the relationship between gaze behavior (e.g., the number of fixations and gaze transitions) and decision-making performance variables (e.g., decision accuracy, decision time, and β) in baseball fielders.

In many previous studies on decision-making and gaze strategies, video clips of gameplay scenes have been presented to participants via a monitor or projector (e.g., Bruce et al., 2012; Kikumasa & Kokubu, 2018; Roca et al., 2011; Vaeyens et al., 2007a, 2007b). While this method offers the advantages of presenting identical stimuli to all participants and ensuring experimental control and reproducibility, there are some concerns that the visual experience differs significantly from that of actual gameplay (Dicks et al., 2010; Farrow & Abernethy, 2003; Mann et al., 2010). Some studies have attempted to address this issue by investigating decision-making and gaze behavior in real-world sports settings (Martell & Vickers, 2004; van Maarseveen et al., 2018). However, in such naturalistic environments, it is difficult to maintain experimental control and reproducibility, as factors such as ball movement and player positioning vary across trials, making it impossible to present identical scenes to each participant. One possible solution to this problem is the use of a head-mounted display (HMD) for stimulus presentation (e.g., DeCouto et al., 2023; Jia et al., 2024; Lee et al., 2024; Panchuk et al., 2018). HMDs enable the presentation of first-person perspectives that more closely resemble real-life conditions, as the visual input is dynamically linked to the wearer's head movements.

Therefore, the aim of this study was to clarify the relationship between gaze behavior and decision-making in baseball players by presenting visual information resembling actual play situations using an HMD, focusing on defensive play situations in baseball, where players are required to make decisions under time constraints. Specifically, we designed a task in which participants judged which base the ball should be thrown to in an infield defensive situation in baseball. Decision accuracy, decision time, and response bias (β) were analyzed as decision-making variables. In addition, we examined gaze behavior in terms of the number of fixations and gaze transition patterns. We hypothesized that (1) decision accuracy would be associated with the number of fixations, with an intermediate number leading to higher accuracy, (2) trials involving gaze directed toward task-irrelevant areas would impair decision-making performance, and (3) gaze transitions would be associated with response bias, and this hypothesis was examined in an exploratory manner given the limited evidence from previous studies.

2. Methods

2.1. Participants

Twenty collegiate baseball fielders participated in the study. Their mean age was 20.7 ± 1.3 years, and their average playing experience was 13.3 ± 1.7 years (Mean ± *SD*). The sample size was determined using a simulation-based a priori power analysis for linear mixed-effects models (LMMs) with the "simr" package in R (Green & MacLeod, 2016). The planned model included random intercepts for participants and stimuli and fixed effects of gaze transition indicators. Assuming medium effect sizes (Cohen's $f^2 = 0.10$–0.15), $\alpha = .05$, and approximately 20–25 trials per participant, the analysis indicated that a minimum of 18–20 participants would be sufficient to achieve $\geq .80$ power for detecting the fixed effects of interest. Accordingly, we recruited 20 participants, which met this requirement. All participants were members of a team competing in the top league of their regional collegiate division. Of the 20 participants, 10 were infielders and 10 were outfielders; none were pitchers or catchers. Prior to the experiment, participants were informed of the study procedures and provided written informed consent. The study protocol was approved by the Ethics Committee of the Institute of Health and Sport Sciences at University of Tsukuba. All procedures were conducted in accordance with the principles of the Declaration of

Helsinki.

2.2. Video Stimuli

In the present study, a defensive scenario was presented in a video from the third baseman's perspective, depicting a one-out situation with a runner on third base. Participants, viewing the scene from the third baseman's point of view, were required to decide which base to throw the ball to—either first base or home plate—within the brief interval between bat–ball contact and the moment the ball was caught. Seven players from the same team appeared as models in the videos: a pitcher, a catcher, a left-handed batter, a third baseman (who was not visible in the videos), and three different third-base runners. Each model was positioned appropriately on the field: the pitcher on the mound, the catcher behind home plate, the batter in the left-handed batter's box, and the third baseman off-screen. The third-base runners appeared one at a time in different video clips (Fig. 1a). The third baseman model wore a 360° camera (Insta360 X3, Arashi Vision Inc.) mounted on the chest using a neck harness, with the lens positioned approximately 150 cm above the ground (Fig. 1b). The camera recorded video at a sampling rate of 30 frames per second.

In all test videos, the left-handed batter, instead of hitting a pitched ball, hit a ball he held in his hand in sync with the pitcher's throwing motion, similar to a fungo hitter, producing a ground ball toward the third baseman. To evaluate whether the runner was safe or out at home plate, the third baseman was instructed to throw the ball to the catcher at home plate as quickly as possible after catching it, and the catcher attempted to tag the runner immediately after catching the ball. The third-base runner took a lead-off in accordance with the pitcher's motion, ran at full speed toward home plate at the moment of bat–ball contact, and slid into home plate. The batter did not run to first base after hitting the ball. Although in real game situations the batter would also run to first base, the run was intentionally omitted because all ground balls in the videos were designed such that an out at first base was always guaranteed. Therefore, the presence of a batter-runner would not affect the decision of whether to throw home. This omission also avoided introducing potential visual distractions, thereby maintaining the primary focus of the task on decisions about whether an out could be made at home plate. Following the recording of the 42 test videos, additional instructional videos were created for participants to watch before the test trials. These videos helped familiarize them with the fielding motion of the third baseman and the base running speed of the runners. Three of these videos showed the third baseman's fielding motion, capturing the sequence from just before the batter's contact to the catcher's reception. An additional three videos depicted the base running motion of the three third-base runners, from their starting movements to their slides into home plate. The third-base runners started approximately 6.5 m from third base, and the time from their start to reaching home plate was 3.3 s for all three runners.

From the 42 test videos, those in which the third baseman's throw bounced before reaching the catcher, or in which the catcher failed to catch the throw, were excluded. Consequently, 24 videos were selected as test videos. For these videos, the time from bat–ball contact to the third baseman's catch was 1.78 ± 0.18 s (range: 1.43–2.07 s). The time from the third baseman's catch to the catcher's catch was 1.21 ± 0.17 s (range: 0.83–1.73 s). Fig. 2 illustrates the test videos used in the experiment. These included 12 "out" trials, in which the catcher tagged the runner 0.13 to 0.54 s before the runner reached home plate, and 12 "safe" trials, in which the runner reached home plate 0.03 to 0.63 s before the tag.

2.3. Procedure

After receiving an explanation of the experimental procedures and tasks, participants stood upright while wearing an eye-tracking head-

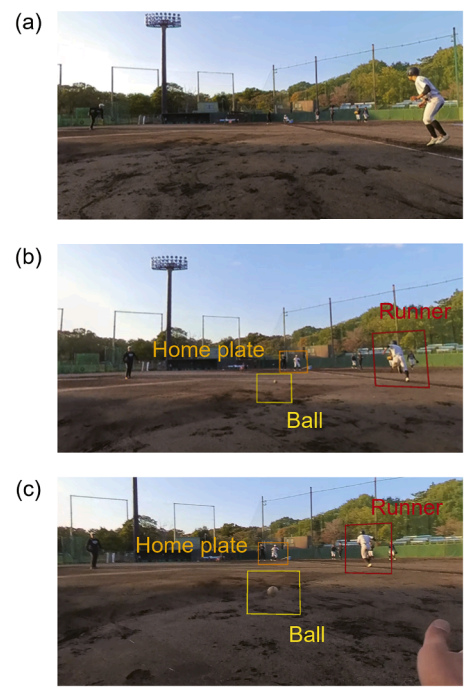

Fig. 2. Example of a test video frame sequence. The images illustrate key moments in the third baseman's decision-making process. (a) The moment of bat–ball contact. (b) The ball and the third baseman approaching each other as the fielder prepares to make the catch. (c) The moment just before the third baseman catches the ball, corresponding to the participant's decision and button press. Note that square frames and texts in this figure were not visible to the participants.

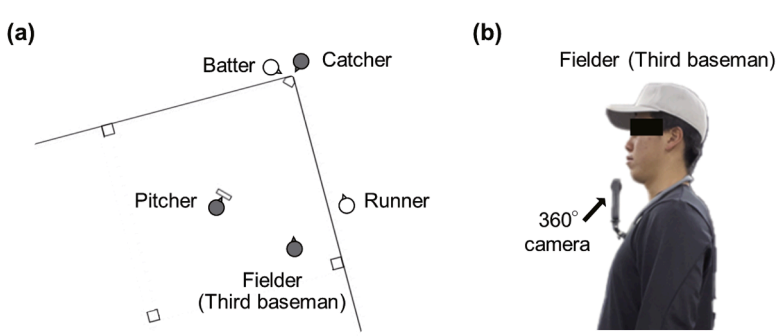

Fig. 1. Experimental setup and camera placement for recording from the third baseman's perspective. (a) Schematic representation of the defensive scenario used in the experiment. The positions of the batter, pitcher, runner, and fielder (third baseman) are indicated in the field diagram. (b) A 360° camera was mounted on the third baseman to record the scene from a first-person perspective.

mounted display (HMD; FOVE 0, FOVE Inc.) and completed the calibration process. This device has been reported to exhibit lower temporal latency in visual stimulus presentation compared to other HMDs (Stein et al., 2021). To ensure that all participants had prior knowledge of the third baseman's and third-base runner's movements, instructional videos were shown prior to the test trials. Specifically, the third baseman's fielding video was presented three times, and the base running video featuring the three third-base runners was shown once. Following this, participants completed four practice trials using videos that were not included in the test set to familiarize themselves with the task.

Subsequently, participants completed 24 test trials, in which each of the selected test videos was presented once in a randomized order. The main task was to decide whether the runner from third base could be put out at home plate. Participants were instructed to press a key on the keyboard to indicate the base to which the third baseman should throw the ball (1 = first base, 4 = home plate) before the moment the third baseman caught the ball. Each test video began 10 seconds prior to bat–ball contact and was occluded at the moment the third baseman caught the ball to prevent feedback-based learning effects. To reduce potential fatigue, a short break was provided after the first 12 trials. The entire procedure, including the instructional videos, practice trials, and test trials, took approximately 25 min to complete.

The participants did not receive any feedback on their success or failure after each trial. Moreover, to focus participants' attention solely on the perceptual-cognitive aspects of the task, no information about the score or inning was provided during the experiment. This was intended to eliminate the potential influence of game-related contextual factors on decision-making. In actual gameplay, players may adjust their decisions based on situational factors such as opting for a safer out at first base when leading, or choosing a riskier play at home plate in a close game. By withholding such information, we aimed to ensure that participants based their decisions purely on the visual cues presented in each trial.

2.4. Dependent variables

2.4.1. Decision accuracy, decision time, and confidence rating

A decision was considered correct when a participant selected home plate in trials where the runner was out at home, and when they selected first base in trials where the runner was safe. Conversely, a decision was considered incorrect when home plate was selected in a "safe" trial or first base was selected in an "out" trial. Additionally, if participants failed to press a key, the response was classified as an omission error. Decision accuracy (%) was calculated by dividing the number of correct responses by the total number of trials ($n = 24$). Decision time (s) was defined as the interval between bat–ball contact and the participant's button press.

In addition, participants rated their confidence in their two-choice decision (i.e., first base or home plate) using a 5-point scale ranging from 1 (not at all confident) to 5 (highly confident). This confidence rating was analyzed as an independent variable to assess whether participants were aware of the accuracy of their own decisions.

2.4.2. Response bias (β)

Decisions were also classified using signal detection theory (Green & Swets, 1966). Table 1 presents the classification scheme used in the present study, based on the study by Kikumasa and Kokubu (2018). In trials where the runner was out at home plate, selecting home plate was categorized as a Hit, and selecting first base as a Miss. When the runner was safe, selecting home plate was classified as a False Alarm (FA), and selecting first base as a Correct Rejection (CR). Likelihood ratio criterion or response bias (β) was calculated based on the criterion (c) and d' using the formula proposed by Macmillan and Creelman (2005):

$$d' = Z(H) - Z(FA) \tag{1}$$

$$c = -0.5 \times [Z(H) + Z(FA)] \tag{2}$$

$$\beta = \exp(d' \times c) \tag{3}$$

where Z(H) and Z(FA) represent the z-scores of hit rate [Hit / (Hit + Miss)] and false alarm rate [FA / (FA + CR)], respectively, and β represents the decision-maker's bias toward taking action based on the perceived likelihood of a signal and the relative costs and benefits of possible outcomes. Specifically, a β greater than 1 suggests a bias toward selecting first base, whereas a β less than 1 suggests a bias toward selecting home plate.

Table 1
Classification of responses based on signal detection theory.

		Participants' decision	
		Home plate	First base
Correct decision	Home plate	Hit	Miss
	First base	False alarm (FA)	Correct rejection (CR)

2.4.3. Gaze behavior indices

Eye-tracking data were analyzed using gaze analysis software (FOVE Gaze Analyzer, FOVE Inc.) to assess gaze behaviors during decision-making. The analyses focused on the time interval between bat–ball contact and the button press.

The FOVE system recorded gaze data at a sampling frequency of 120 Hz with a reported tracking accuracy of 1.15° of visual angle. Given that this accuracy is comparable to that of eye-tracking systems used in previous research, we adopted the conventional definition of a fixation as maintaining gaze within a visual angle of 3 deg for at least 100 ms (Vickers, 2007, p. 19). The number of fixations per trial was calculated for the specified time interval.

Gaze transition patterns were classified using an Area of Interest (AOI) approach. The AOIs were defined as the ball (B), the runner (R), and the home plate (H). Viewing time on the ball was calculated based on the duration of gaze fixations within that area, and the proportion of viewing time spent on the ball was expressed as a percentage of the total fixation time. Fixations outside these predefined AOIs were categorized as other areas (O). As noted above (see *Video Stimuli*), the batter did not run to first base in the video stimuli, and the batter-runner was not considered relevant to the participants' decision-making process. Therefore, the batter-runner was not included as an AOI. The number of transitions between AOIs per trial was also counted as an index of visual search behavior. For example, a pattern labeled "BH" indicates a gaze transition from the ball to the home plate and was counted as one transition, whereas "B" indicates gaze remained on the ball AOI throughout the trial and was counted as no transition.

2.5. Data analysis

Data from all participants were included in the analysis. To assess whether the overall decision accuracy was significantly above the chance level (50%), a binomial test was conducted on the number of correct responses out of the total trials ($n = 480$). Trials classified as omission errors were excluded from subsequent analyses.

The normality of residuals was assessed using the Shapiro–Wilk test. The results indicated that the residuals from the regression models predicting β deviated from normality. Therefore, robust regression was employed for the analyses involving β. This approach reduces the influence of outliers, providing more reliable estimates than ordinary least squares regression. For the other variables, standard regression and mean-comparison tests were conducted, given that parametric tests are generally robust to minor violations of the normality assumption (Norman, 2010).

Mean decision time and confidence rating were calculated separately for correct and incorrect trials for each participant. To compare correct

and incorrect trials for each of the four dependent variables, paired *t*-tests were conducted. Effect sizes were calculated using Cohen's *d*.

To examine the relationship between the number of fixations per trial and decision accuracy, polynomial regression analyses were conducted. Both linear and quadratic models were fitted to the data, and model fit was evaluated based on the coefficient of determination (R^2) and statistical significance. The analysis aimed to determine whether a non-linear (quadratic) trend better explained variations in decision accuracy as a function of fixation frequency.

To examine whether decision accuracy differed depending on the presence or absence of gaze at each AOI, chi-square tests were conducted by comparing trials with and without gaze at each AOI. Effect sizes were calculated using phi coefficients (φ).

To examine whether decision time differed depending on the presence or absence of gaze at each AOI, LMM analyses were used. Although *t*-tests were used in other comparisons, they were not suitable here due to the absence of gaze at certain AOIs in some participants, resulting in missing data for within-subject comparisons. LMMs were therefore selected to handle the unbalanced data and to model both fixed and random effects simultaneously. The primary model included fixed effects of gaze at the ball (GazeB), home plate (GazeH), runner (GazeR), and other areas (GazeO), along with random intercepts for participants (ParticipantID) and stimuli (StimID) to account for participant- and item-level variability. The model formula was as follows:

$$\text{Decision time} \sim \text{GazeB} + \text{GazeH} + \text{GazeR} + \text{GazeO} + (1 \mid \text{ParticipantID}) + (1 \mid \text{StimID}) \quad (4)$$

Here, all fixed-effect variables (GazeB, GazeH, GazeR, and GazeO) were binary indicators denoting the absence (0) or presence (1) of a gaze at each AOI during a trial. ParticipantID and StimID were treated as categorical grouping variables for random intercepts.

Robust regression analyses were conducted using M-estimation with Huber's ψ-function to examine the relationships between response bias (β) and gaze behavior indices such as the proportion of viewing time spent on the ball and the number of transitions between AOIs.

All statistical analyses were performed using SPSS ver. 29 (IBM Corp.) except for LMM analyses using the lmer function from the lme4 package in R ver. 4.4.2 and robust regression analyses using the rlm function from the MASS package in R ver. 4.4.2. The significance level was set at $\alpha = .05$, and all *p*-values were reported accordingly.

3. Results

3.1. Decision accuracy

The mean decision accuracy across all participants was 66.7 ± 5.1%. Of the 480 total trials, 320 were correct trials, 146 were incorrect trials, and 14 were omission error trials. A binomial test showed that the decision accuracy was significantly above the chance level (50%), $p < .001$. Omission errors were treated as incorrect responses for the decision accuracy analysis, but because no response time could be recorded and gaze behavior was defined relative to the button press, they were excluded from further analyses. Accordingly, subsequent analyses were conducted on the remaining 466 trials in which the response button was pressed.

3.2. Decision time and confidence rating

Table 2 shows means and *SDs* of decision time and confidence rating for correct and incorrect decisions. Paired *t*-tests showed no significant difference in the decision time between correct (1.53 ± 0.19 s) and incorrect (1.52 ± 0.16 s) decisions [$t(19) = 0.217$, $p = .830$, $d = 0.049$]. In contrast, the confidence rating was significantly higher for correct decisions (3.87 ± 0.34) than for incorrect decisions (2.58 ± 0.50) [$t(19) = 11.952$, $p < .001$, $d = 2.673$].

Table 2
Means and standard deviations of decision time and confidence rating for correct and incorrect decisions.

Variables		Correct	Incorrect
Decision time (s)	*M*	1.53	1.52
	SD	0.19	0.16
Confidence rating (1-5)	*M*	3.87***	2.58
	SD	0.34	0.50

*** Significant difference between correct and incorrect decisions ($p < .001$).

3.3. Relationship between the number of fixations and decision accuracy

The mean number of fixations per trial across all participants was 2.69 ± 0.43. Polynomial regression analysis to examine the relationship between the number of fixations per trial and decision accuracy demonstrated that the quadratic model ($y = -12.74x^2 + 71.48x - 31.23$, $R^2 = .389$, $p = .015$) provided a significantly better fit to the data than the linear model ($y = 3.78x + 56.52$, $R^2 = .102$, $p = .170$), indicating an inverted U-shaped relationship between the number of fixations per trial and decision accuracy (Fig. 3).

3.4. Gaze transition patterns

Table 3 summarizes the gaze transition patterns observed in the present study. Participants predominantly fixated on the ball first, with this pattern occurring in 88.6% of trials (413 out of 466 trials). Among these, the most frequent pattern was the single gaze at the ball (B), which accounted for 41.8% of all trials and resulted in a decision accuracy of 72.3%. In contrast, patterns beginning from the home plate, other areas, and runner were less common (5.8%, 5.4%, and 0.2%, respectively).

3.5. Effect of gaze at each AOI on decision accuracy

Table 4 shows the effect of gaze at each AOI on decision accuracy. Chi-square tests revealed that trials in which participants gazed at other areas (O) had significantly lower decision accuracy (57.1%; 36 correct out of 63 trials) compared to trials without such gaze (70.5%; 284 correct out of 403 trials) [$\chi^2(1) = 4.50$, $p = .034$, $\varphi = .10$]. In contrast, decision accuracy in trials with gaze at the ball (B) (69.6%: 309 correct out of 444 trials) did not significantly differ from that in trials without such gaze (50.0%: 11 correct out of 22 trials) [$\chi^2(1) = 3.74$, $p = .053$, $\varphi =$

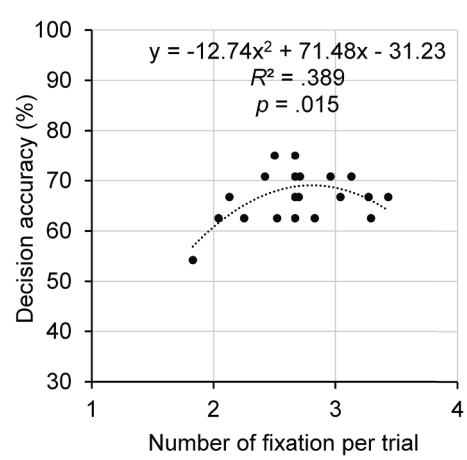

Fig. 3. Relationship between the number of fixations per trial and decision accuracy. A second-order polynomial regression analysis revealed a significant quadratic relationship between the number of fixations per trial and the decision accuracy (%). The regression model is shown in the figure, with the best-fit curve, coefficient of determination ($R^2 = .389$), and *p*-value ($p = .015$) indicating a significant effect.

Table 3
Distribution of gaze transition patterns categorized by initial fixation location.

Patterns	*N*	Ratio (%)	Correct (*n*)	Incorrect (*n*)	Decision accuracy (%)
Patterns beginning from the ball (B)					
B	195	41.8	141	54	72.3
BH	121	26.0	79	42	65.3
BHB	3	0.6	2	1	66.7
BHBRH	1	0.2	1	0	100.0
BHO	5	1.1	2	3	40.0
BHOH	1	0.2	1	0	100.0
BHOR	1	0.2	1	0	100.0
BHR	13	2.8	10	3	76.9
BHRH	1	0.2	1	0	100.0
BO	16	3.4	9	7	56.3
BOBH	2	0.4	2	0	100.0
BOH	5	1.1	3	2	60.0
BR	27	5.8	20	7	74.1
BRB	1	0.2	1	0	100.0
BRBR	2	0.4	2	0	100.0
BRH	17	3.6	12	5	70.6
BRHR	1	0.2	1	0	100.0
BRO	1	0.2	1	0	100.0
Total	413	88.6	289	124	70.0
Patterns beginning from the home plate (H)					
H	9	1.9	6	3	66.7
HB	5	1.1	4	1	80.0
HBH	2	0.4	1	1	50.0
HBO	2	0.4	0	2	0.0
HBR	1	0.2	0	1	0.0
HBRH	1	0.2	1	0	100.0
HO	3	0.6	0	3	0.0
HOH	2	0.4	1	1	50.0
HR	2	0.4	1	1	50.0
Total	27	5.8	14	13	51.9
Patterns beginning from other areas (O)					
O	5	1.1	2	3	40.0
OB	9	1.9	5	4	55.6
OBH	4	0.9	3	1	75.0
OBO	1	0.2	1	0	100.0
OBR	1	0.2	1	0	100.0
OBRH	3	0.6	2	1	66.7
OH	1	0.2	1	0	100.0
OHB	1	0.2	1	0	100.0
Total	25	5.4	16	9	64.0
Pattern beginning from the runner (R)					
RBH	1	0.2	1	0	100.0
Total	1	0.2	1	0	100.0

Note. B = ball; H = home plate; R = runner; O = other areas. For example, "BH" indicates gaze transition from the ball to the home plate.

Table 4
Effect of gaze at each area of interest on decision accuracy.

		Correct (*n*)	Incorrect (*n*)	Decision accuracy (%)
Gaze at ball (B)				
	Without	11	11	50.0
	With	309	135	69.6
Gaze at home plate (H)				
	Without	183	75	70.9
	With	137	71	65.9
Gaze at runner (R)				
	Without	264	128	67.3
	With	56	18	75.7
Gaze at other areas (O)				
	Without	284	119	70.5
	With	36	27	57.1*

* Significant difference between trials without and with gaze ($p < .05$).

.09]. Similarly, no significant differences in decision accuracy were found between trials with and without gaze at the home plate (H) [$\chi^2(1) = 1.37, p = .241, \varphi = .05$], or to the runner (R) [$\chi^2(1) = 2.01, p = .157, \varphi = .06$].

3.6. Effect of gaze at each AOI on decision time

The LMM analysis revealed significant effects of gaze at the home plate [GazeH: $\beta = 0.154$, $SE = 0.025$, $t(454) = 6.17$, $p < .001$], the runner [GazeR: $\beta = 0.113$, $SE = 0.034$, $t(448) = 3.36$, $p < .001$], and other areas [GazeO: $\beta = 0.129$, $SE = 0.034$, $t(443) = 3.77$, $p < .001$], indicating that gaze at these areas were associated with significantly longer decision times by 154 ms, 113 ms, and 129 ms, respectively. In contrast, gaze at the ball [GazeB: $\beta = 0.100$, $SE = 0.055$, $t(441) = 1.81$, $p = .071$] did not show a statistically significant effect on decision time (Fig. 4).

3.7. Relationship between response bias and gaze behavior indices

For the variables related to signal detection theory, the mean β was 1.14 ± 0.72. Regarding gaze behavior indices, the mean proportions of viewing time spent on the ball was $54.0 \pm 15.2\%$. The mean number of transitions between AOIs was 0.74 ± 0.44.

Robust regression analyses revealed a significant negative relationship between β and the proportion of viewing time spent on the ball ($b = -1.24$, $SE = 0.60$, $z = -2.07$, $p = .038$), indicating that participants who allocated more gaze time to the ball tended to select home plate more frequently (Fig. 5a). In addition, a significant positive relationship was observed between β and the number of transitions between AOIs ($b = 0.54$, $SE = 0.17$, $z = 3.24$, $p = .001$), indicating that participants who made more gaze transitions were more likely to select first base (Fig. 5b).

4. Discussion

The present study investigated the relationship between gaze behavior and decision-making performance in a simulated defensive baseball task. The findings provide insights into how gaze behavior is associated with decision accuracy and decision time under time-constrained conditions.

Participants' decision accuracy was significantly above chance level, with an average accuracy of 66.7%. This indicates that they were able to make informed decisions based on available visual cues rather than responding randomly. The ability to consistently identify the appropriate base to throw to, even in a controlled experimental setting, reflects the capacity to extract relevant information and apply meaningful strategies, possibly shaped by prior experience or task familiarity (Williams & Ericsson, 2005; Vaeyens et al., 2007b). The results also suggest that the HMD-based stimuli provided a controlled and useful approach for examining decision-making processes in baseball defensive scenarios, although conclusions regarding ecological validity remain limited without direct comparison to real-life performance.

There was no significant difference in decision time between correct and incorrect trials, suggesting that participants did not demonstrate a speed–accuracy tradeoff when deciding whether to throw to first base or home plate. In other words, taking more time did not lead to more accurate decisions, nor did faster responses result in more errors. Confidence ratings were significantly higher for correct decisions, indicating that participants were aware of the accuracy of their responses even without trial-by-trial feedback. Previous studies using random dot motion stimuli have shown that a higher degree of certainty is generally associated with shorter decision times (Kiani et al., 2014). However, this pattern was not observed in the present study. One possible explanation is that the task did not require participants to respond as quickly as possible. Instead, they were instructed to make their decision within the time available before the fielder caught the ball. This design allowed for flexible response timing and likely reduced the pressure to prioritize speed. In fact, studies conducted in sports contexts have reported that although skilled athletes tend to make more accurate decisions than less skilled individuals, they do not necessarily respond more quickly

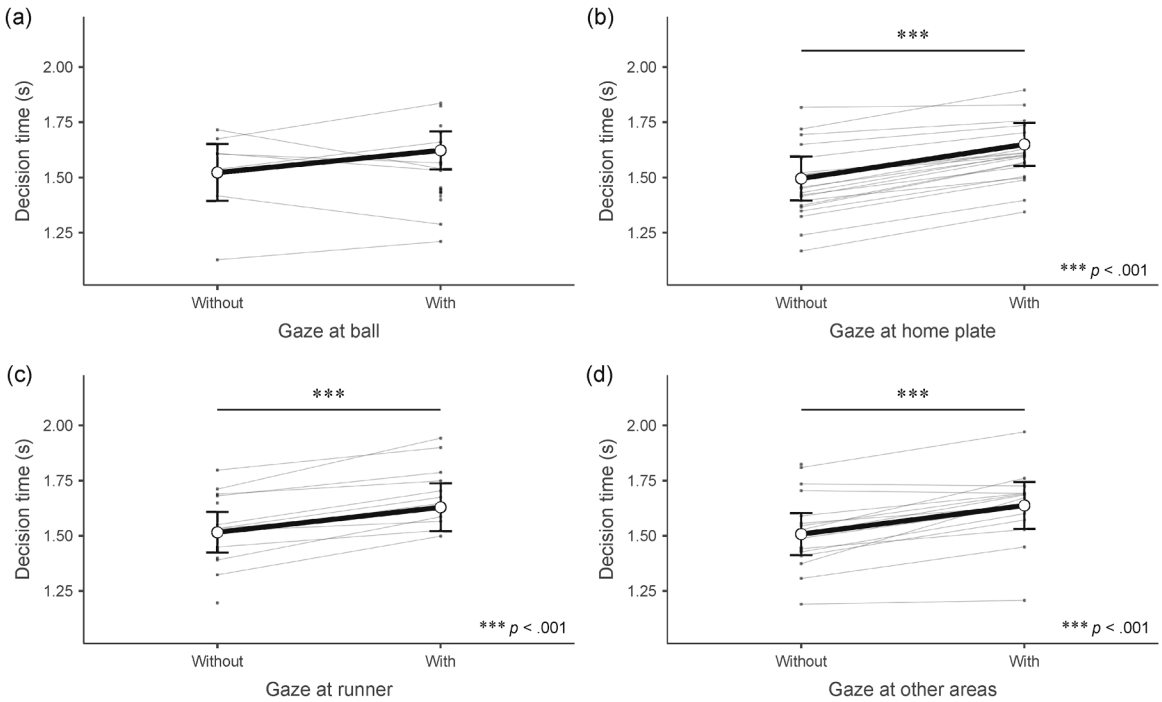

Fig. 4. Effect of gaze at each area of interest on decision time. Fixed effects of gaze at the (a) ball, (b) home plate, (c) runner, and (d) other areas are shown. Decision times were compared between trials with and without gaze at each area, using a linear mixed-effects model with random intercepts for participants and stimuli. Individual random effects are represented by participant-level data (thin lines). Error bars represent standard errors of the mean. *** $p < .001$.

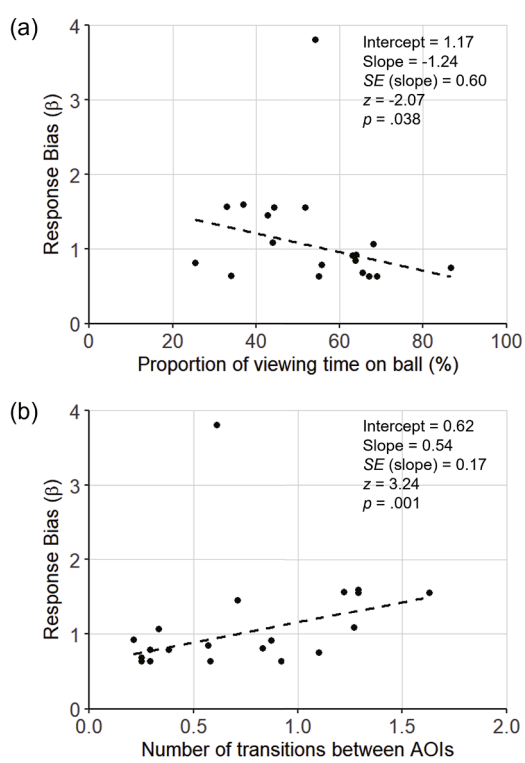

Fig. 5. Relationships between response bias (β) and gaze behavior indices using robust linear regression. (a) A significant negative association was observed between response bias and the proportion of viewing time spent on the ball ($p = .038$). (b) A significant positive association was observed between response bias and the number of transitions between areas of interest (AOIs) ($p = .001$). Robust linear regression (M-estimation with Huber's ψ-function) was used for both analyses to mitigate the influence of outliers and non-normal residuals. Dashed lines show robust fitted trends.

(Kikumasa & Kokubu, 2018; Ripoll et al., 1995; Savelsbergh et al., 2002; Vansteenkiste et al., 2014). The present findings are consistent with these studies, suggesting that correct decisions in this task were not simply the result of rapid responses but were instead reflective of deliberate and accurate processing within the available time frame.

The results demonstrated a quadratic relationship between the number of fixations per trial and decision accuracy, forming an inverted U-shaped pattern with peak accuracy observed at approximately 2–3 fixations within a 1.5-s window (Fig. 3). Many previous studies have shown that skilled players tend to make fewer fixations than novices (Mann et al., 2007 for review; Piras et al., 2014). On the other hand, in soccer penalty kicks, for example, goalkeepers tend to look at the ball for too long during trials in which they fail to make a save (Piras & Vickers, 2011). Additionally, previous studies have suggested the importance of visual pivots in interpersonal sports (Kato, 2020; Ripoll et al., 1995; Savelsbergh et al., 2002). These discrepancies across studies may reflect differences in both skill level and task characteristics (Ziv, 2025). For instance, experts in relatively static tasks such as aiming may benefit from fewer and longer fixations, whereas athletes in dynamic contexts such as soccer or baseball may require more frequent fixations to integrate rapidly changing cues. In the present study, the intermediate-level sample and the time-constrained defensive scenario may have jointly contributed to the emergence of an optimal fixation range rather than a simple linear trend.

Consistent with our original hypothesis, the results indicate that decision accuracy is related to the number of fixations. This suggests that both insufficient and excessive visual scanning can impair decision-making, and that a moderate level of visual sampling (i.e., around one fixation every 0.5 s) may strike an optimal balance between information gathering and cognitive processing. These findings align with previous research emphasizing the importance of efficient, rather than exhaustive, visual search strategies in sports contexts (Mann et al., 2007; Williams et al., 1999). Moreover, Roca et al. (2011) showed that skilled participants employed a greater number of fixations of shorter duration, directed at more informative and spatially disparate locations, in comparison with less skilled participants, indicating that fixation transitions

were more efficient and allowed for the processing of more information in less time. This observation is partly consistent with the present findings. Such interpretations are consistent with visual search theories, which propose that expert performance is supported by search strategies that balance information extraction and processing efficiency (Ziv, 2025). This theoretical perspective provides further support for our finding that a moderate number of fixations represents an optimal strategy for decision-making in time-constrained defensive scenarios. From an applied perspective, this implies that an optimal allocation of attention, which is neither overly narrow nor excessively dispersed, enhances decision accuracy by ensuring that critical information is extracted while avoiding cognitive overload. Future studies should examine whether similar fixation patterns are observed in real-game scenarios to further elucidate the role of gaze behavior in expert-level baseball fielding.

The findings of gaze transition patterns revealed that fielders primarily fixated on the ball first in most trials (Table 3). Additionally, trials where participants fixated on unrelated areas (O) were less frequent and associated with lower decision accuracy (Table 4). These results are in line with our hypothesis and suggest that successful defensive decision-making relies on initially using the ball as a visual anchor and information-rich cue, which is consistent with earlier research (Kikumasa & Kokubu, 2018). Gaze at less informative areas was associated with significantly lower decision accuracy, suggesting that attending to irrelevant regions may hinder decision-making performance. In the context of the present study, fielders should prioritize tracking the ball and avoid excessive fixations on unrelated areas.

The linear mixed-effects model showed that gaze at the home plate, runner, and other areas were associated with significantly longer decision times (Fig. 4). Specifically, trials involving gaze at these areas resulted in delays of approximately 100 to 150 ms compared to trials without such gaze. This finding suggests that integrating information across multiple areas comes at a time cost. It is important to note that there was no difference in decision accuracy depending on whether gaze was directed to the home or runner. In contrast, gaze at the ball did not significantly affect decision time, underscoring its role as a primary and default focal point during the task.

Response bias (β) was found to be significantly related to individual gaze behavior. Participants who spent more time fixating on the ball tended to exhibit a bias toward selecting home plate, while those who made more frequent gaze transitions showed a bias toward selecting first base (Fig. 5). The results were consistent with our exploratory hypothesis regarding the link between gaze transitions and response bias. One possible explanation for this relationship is that fixating on the ball allows fielders to extract time-to-contact (TTC) information of the ball, which helps them assess whether they have enough time to throw to home plate to achieve an out. In fact, it has been shown that the accuracy of TTC estimation is higher for stimuli presented in central vision than for those in peripheral vision (Regan & Vincent, 1995). Since deciding to throw to home plate involves greater time pressure than deciding to throw to first base, fielders focusing on the ball may prioritize TTC information to rapidly decide whether a play at home is viable. In contrast, when fielders determine that an out at home is unlikely and opt to throw to first base instead, they are afforded more time to visually assess other cues, such as the runner's movement. However, it should be noted that information about the batter-runner was not included in the present task, which may have limited the ecological validity of these observations. This reduced time pressure may lower the need to maintain fixation on the ball, resulting in a broader gaze distribution and more frequent gaze transitions—patterns indicative of a more conservative decision-making strategy. These findings highlight the importance of gaze allocation in defensive decision-making, suggesting that fielders adjust their visual strategies according to their intended play. Fixating on the ball thus appears to support time-critical decisions, while a wider distribution of gaze may facilitate more deliberative choices when time permits. Understanding these gaze patterns and their relationship to decision-making tendencies could inform perceptual-cognitive training programs aimed at improving fielding decisions under pressure.

The findings of this study suggest that visual search strategies play a crucial role in decision-making during defensive situations in baseball, particularly under strict time constraints. Building on these findings, practical applications can be considered. Although the present task did not fully replicate real game situations because the batter did not run to first base and participants were not required to actually make the throw, the use of video-based scenarios would still provide a useful approach for examining and training gaze strategies. Future tasks that include more realistic elements, such as a batter running or requiring an actual throwing action, may further enhance their effectiveness as training. In particular, the use of HMDs equipped with eye-tracking capabilities allows for immediate feedback on gaze behavior, enabling athletes to adjust their visual strategies in real time. These applications highlight the potential benefits of integrating perceptual-cognitive training into baseball skill development programs to enhance defensive performance and reduce decision errors during high-pressure plays.

Overall, the present findings highlight the systematic and nuanced role of gaze patterns in sport-specific decision-making. The interaction between gaze strategies and decision-related factors such as response bias provides deeper insight into the perceptual-cognitive processes underlying rapid decision-making in dynamic sports contexts. Importantly, while previous studies have examined gaze behavior in batters (e.g., Kishita et al., 2020; Takeuchi & Inomata, 2009; Toole & Fogt, 2021) and catchers (Kikumasa & Kokubu, 2018), the present study is the first to investigate fielders' defensive decision-making using a HMD system that simulates real-game perspectives. Future research should consider manipulating task constraints and contextual information or comparing participants with varying levels of expertise to further elucidate how perceptual-cognitive skills develop and adapt in high-performance environments (Broadbent et al., 2015; Roca et al., 2012).

One limitation of this study is that the HMD used in the present experiment had a field of view of 100°, which is narrower than the actual human visual field (approximately 180–190°). As a result, participants' natural peripheral vision—particularly far peripheral vision—may not have been fully utilized. For example, in real-world scenarios, players may detect a base runner through peripheral vision rather than direct fixation. However, given that the experimental task in the present task did not require a wide field of view, this limitation is unlikely to have significantly influenced the results. Future studies should investigate how far peripheral vision contributes to decision-making in baseball fielding situations.

Another limitation of the current study is that perception and action were not coupled. Participants made decisions based on video stimuli without physically executing the throwing motion. Prior research has shown that gaze behavior can differ when perceptual information is directly linked to motor execution, as compared to video-based simulation tasks (Dicks et al., 2010; Mann et al., 2010). Thus, while the current design allowed for precise control over visual stimuli, it may not fully reflect the perceptual-motor integration that occurs in real-world gameplay. Future research should consider designs that incorporate simulated throwing actions by baseball fielders under the HMD conditions (e.g., Lee et al., 2024), in order to better capture how decision-making is influenced by throwing ability.

5. Conclusions

This study investigated the relationship between gaze behavior and decision-making in simulated baseball fielding situations. Decision accuracy was highest with an intermediate number of fixations per trial, while too few or too many fixations reduced accuracy. Gaze patterns revealed that players typically fixated on the ball at the start of each trial, while gaze directed to home plate, runner, or irrelevant areas prolonged decision time, and gaze directed to irrelevant areas reduced accuracy. Response bias (β) correlated with gaze behavior: longer ball

fixations were associated with assertive decisions (throwing to home plate), whereas frequent gaze transitions predicted more conservative choices (throwing to first base). These findings suggest that both the stability and allocation of gaze play key roles in shaping decision-making under time constraints. Training programs targeting efficient gaze strategies may enhance fielding performance and tactical decision-making in competitive baseball.

Author contributions

Masahiro Kokubu: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing - original draft, Writing – review & editing. **Koki Shojima:** Data curation, Formal analysis, Investigation, Visualization, Writing - original draft. **Syunpei Kikumasa:** Formal analysis, Validation, Writing - original draft, Writing – review & editing. **Yuta Oki:** Validation, Writing – review & editing. **Takumi Mieda:** Validation, Writing – review & editing.

Data availability statement

The datasets for this study can be found on OSF at https://osf.io/fnxku/.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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